

III Laplace Transformation:

* Transformation: Change one state to another

* Integral Transformation

* Operator function is called Kernel (Function)

* Transforms ^{space} variable to another constant term using Kernel

$$F(s) = \int_a^b K(s, t) f(t) dt, \quad -\infty \leq a, < b \leq \infty$$

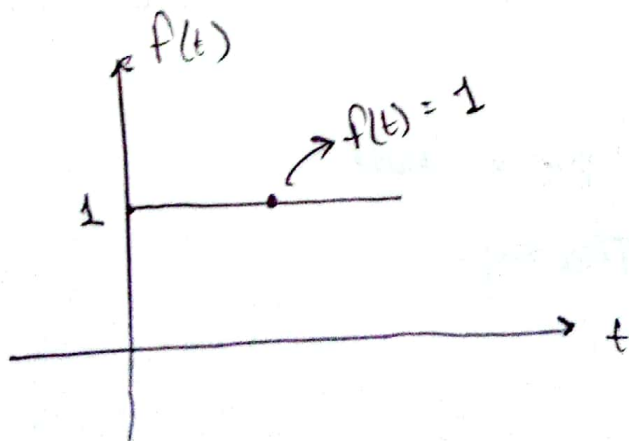
$\downarrow$ Transformation $\nwarrow$ Kernel $\searrow$ function of t

* NOT Laplace

* Let, f be a function defined for $t \geq 0$ and satisfies certain conditions to be named later.

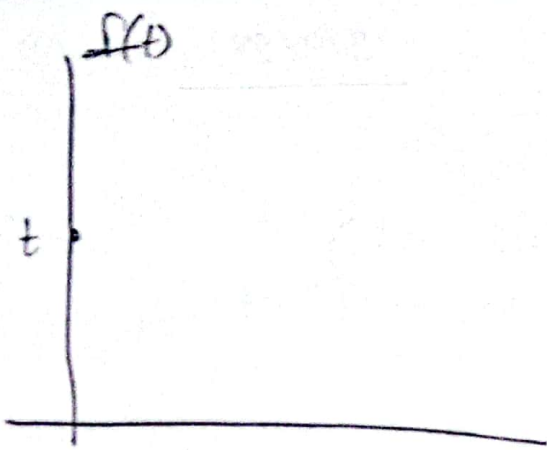
$\therefore$ The Laplace Transformation of f is defined as:

$$L\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$



$$L\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

$$\begin{aligned} &= \int_0^{\infty} e^{-st} \cdot 1 dt \\ &= \left[-\frac{1}{s} e^{-st} \right]_0^{\infty} \\ &= -\frac{1}{s} \left(e^{-s\infty} - 1 \right) \\ &= \frac{1}{s} \end{aligned}$$



$$\int_0^{\infty} e^{-st} t dt$$

$$= \int_0^{\infty} e^{-st} t \cdot \frac{d(e^{-st})}{-st e^{-st}}$$

$$= -\frac{1}{s}$$

$$= \int_0^{\infty} t \times \frac{e^{-st}}{-s} - \int_0^{\infty} 1 \cdot \frac{e^{-st}}{-s} dt$$

$$= 0 - 0 + \left[\frac{e^{-st}}{-s^2} \right]_0^{\infty}$$

$$= \frac{1}{s^2}$$

$$\frac{d e^{-st}}{dt}$$

$$dv = e^{-st}$$

$$\therefore v = -\frac{1}{s} e^{-st}$$

$$u = t$$

$$\therefore du = dt$$

$$\int u dv = \int \frac{d(u)}{dx}$$

$$\int u v dx$$

$$= u \int v dx - \int \frac{d}{dx} u \int v dx$$

H.W

$t^2, t^3, t^n, \sin t, \cos t$

$$(c-3)(w-3)$$

21/04/24

$$\text{Find } f(t) = e^{at} \text{ for } t \geq 0$$

$$\text{Sol}^n: \text{Let } L\{e^{at}\} = \int_0^{\infty} e^{-st} \cdot e^{at} dt$$

$$= \int_0^{\infty} e^{t(a-s)} dt$$

$$= \left[\frac{e^{t(a-s)}}{a-s} \right]_0^{\infty}$$

$$= \left(0 - \frac{e^{-\infty}}{a-s} - \frac{e^{0(a-s)}}{a-s} \right)$$

$$= 0 - \frac{1}{a-s}$$

$$= \frac{1}{s-a}$$

$$\therefore L\{e^{-2t}\} = \frac{1}{s+2}$$

$$\boxed{f(t) = \sin at}$$

Euler Formula :

$$\begin{cases} e^{i\theta} = \cos\theta + i\sin\theta \\ e^{-i\theta} = \cos\theta - i\sin\theta \end{cases} \begin{cases} (+) \Rightarrow e^{i\theta} + e^{-i\theta} = 2\cos\theta \\ (-) \Rightarrow e^{i\theta} - e^{-i\theta} = 2i\sin\theta \end{cases}$$

$$\mathcal{L}\{\sin at\} = \int_0^{\infty} e^{-st} \cdot \sin at \, dt$$

$$= \int_0^{\infty} e^{-st} \cdot \frac{1}{2i} (e^{iat} - e^{-iat}) \, dt$$

$$= \frac{1}{2i} \int_0^{\infty} e^{(ia-s)t} - e^{-(ia+s)t} \, dt$$

$$= \frac{1}{2i} \left[\frac{e^{(ia-s)t}}{ia-s} - \frac{e^{-(ia+s)t}}{-(ia+s)} \right]_0^{\infty}$$

$$= \frac{1}{2i} \left[-\frac{1}{ia-s} - \frac{1}{ia+s} \right]$$

$$= \frac{-1}{2i} \times \frac{ia-s+ia+s}{(ia)^2 - (s)^2}$$

$$= \frac{1}{2i} \times \frac{2i \times a}{s^2 + a^2}$$

$$= \frac{a}{s^2 + a^2}$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

$$\oplus e^{i\theta} + e^{-i\theta} = 2 \cos \theta$$

$$\ominus e^{i\theta} - e^{-i\theta} = 2i \sin \theta$$

hyperbolic function

↑ called

(complex conjugate) $e^{i\theta} + e^{-i\theta} = 2 \cosh \theta$

$$\mathcal{L}\{\cosh at\} = \int_0^{\infty} e^{-st} \cosh at \, dt$$

$$= \int_0^{\infty} e^{-st} \times \frac{1}{2} (e^{iat} - e^{-iat}) \, dt$$

$$= \frac{1}{2} \int_0^{\infty} e^{t(ia-s)} - e^{t(-s-ia)} \, dt$$

$$= \frac{1}{2} \left[-\frac{e^{t(ia-s)}}{ia-s} - \frac{e^{t(-s-ia)}}{-s-ia} \right]_0^{\infty}$$

$$= \frac{1}{2} \left[-\frac{1}{ia-s} - \frac{1}{-s-ia} \right]$$

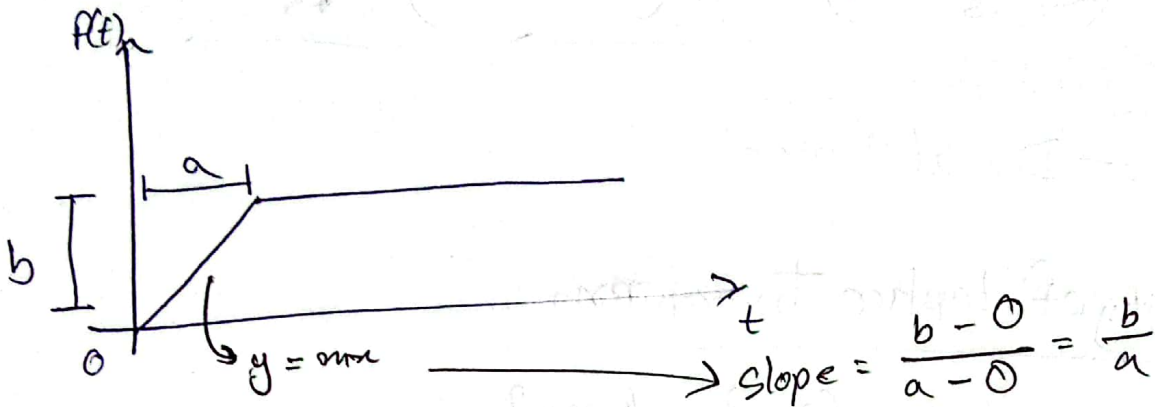
$$= \frac{1}{2} \left(\frac{-ia-s - (-s-ia)}{-a^2 - s^2} \right)$$

$$= \frac{-2ia}{2(-a^2 - s^2)} = \frac{ia}{s^2 + a^2}$$

H.W: $L \{ \sinh at \}$
 $L \{ \cosh at \}$

Example 12: Perform the Laplace transformation on the ramp illustrated below:

1. Find mathematical form
2. Do Laplace



From the graph, we have,

$$f(t) = \begin{cases} \frac{b}{a}t & ; t \leq a \\ b & ; t \geq a \end{cases} \rightarrow \text{piecewise function}$$

$$\therefore L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$= \int_0^a e^{-st} f(t) dt + \int_a^{\infty} e^{-st} f(t) dt$$

$$= \int_0^a e^{-st} \frac{b}{a} dt + \int_a^{\infty} e^{-st} b dt$$

$$= \frac{b}{a} \int_0^a t e^{-st} dt + b \int_a^{\infty} e^{-st} dt$$

$$= \frac{b}{a} \left[\frac{t}{s} \right]$$

$$= \frac{b}{a} \left(\left[t \cdot -\frac{1}{s} e^{-st} \right]_0^a - \int_0^a -\frac{1}{s} e^{-st} dt \right) + b \left[\frac{e^{-st}}{-s} \right]_a^\infty$$

$$\begin{aligned} u &= t \\ u &= t \\ du &= dt \end{aligned} \quad \begin{aligned} v &= e^{-st} \\ v &= \frac{1}{s} e^{-st} \end{aligned}$$

$$= \frac{b}{a} \left(a \left(-\frac{1}{s} e^{-as} \right) + \frac{1}{s^2} (e^{-as} - 1) \right) + \frac{b}{s} e^{-as}$$

~~Q~~ = Do at home

Linearity of Laplace transform:

$$\mathcal{L}\{f(t) + g(t)\} = \mathcal{L}\{f(t)\} + \mathcal{L}\{g(t)\}$$

$$\# f(t) = \underbrace{5e^{2t}}_{\mathcal{L}\{f(t)\}} + \underbrace{3\sin(4t)}_{\mathcal{L}\{g(t)\}}, \text{ for } t \geq 0$$

Do @ home

C-5(w-4)

28/04/24

Laplace Transformation

Properties:

- * Linearity
- * Change of scale
- * Time shift ; $[x \text{ is time ; Domain is time}]$
- * "frequency" on s-plane shift ; $[t \rightarrow et // at // whatever t]$
- * Multiplication by t^n
- * Integration
- * Differentiation

P.T.O

Linearity

$$\mathcal{L}\{c_1 f_1(t) + c_2 f_2(t)\} = c_1 F_1(s) + c_2 F_2(s)$$

*check prev class

*check slide

Change of Scale

$$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$$

$$\rightarrow \mathcal{L}\{\sin t\} = \frac{1}{s^2 + a^2}$$

$$\therefore \mathcal{L}\{\sin at\} = ?$$

Proof:

$$\mathcal{L}\{f(at)\} = \int_0^{\infty} f(at) e^{-st} dt$$

$$\text{let } u = at, \quad t = \frac{u}{a}$$

$$dt = \frac{1}{a} du$$

$$\therefore \frac{1}{a} \int_0^{\infty} f(u) e^{-\left(\frac{s}{a}\right)u} du$$

→ still ∞

$$= \frac{1}{a} F\left(\frac{s}{a}\right)$$

Example:

*fix and do slide

Time Shift

unit step function

$$L\{f(t-t_0)u(t-t_0)\} = e^{-st_0} F(s)$$

has only 2 outputs (0 or 1)

$$u(t-t_0) = \begin{cases} 1; & t \geq t_0 \\ 0; & t < t_0 \end{cases}$$

Proof:

$\therefore$ this will be 0 or 1

$$L\{f(t-t_0)u(t-t_0)\}$$

$$= \int_0^{\infty} f(t-t_0)u(t-t_0)e^{-st} dt$$

$$= \int_{t_0}^{\infty} f(t-t_0)e^{-st} dt$$

Let,

$$u = t - t_0, \quad t = u + t_0$$

$$= \int_0^{\infty} f(u)e^{-s(u+t_0)} du$$

$$= e^{-st_0} \int_0^{\infty} f(u)e^{-su} du = e^{-st_0} F(s)$$

Example

$$L\{e^{-a(t-10)}u(t-10)\}$$

$$= \frac{e^{-10s}}{s+a}; \quad L(e^{-at}) = \frac{1}{s+a}$$

III S-Plane (Frequency) Shift

$$\mathcal{L}\{e^{-at} f(t)\} = F(s+a)$$

Proof:

$$\mathcal{L}\{e^{-at} f(t)\}$$

$$= \int_0^{\infty} e^{-at} f(t) e^{-st} dt$$

$$= \int_0^{\infty} f(t) e^{-(s+a)t} dt$$

$$= F(s+a)$$

Example:

III Multiplication by t^n

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

Example:

Proof:

$$\mathcal{L}\{t^n f(t)\}$$

$$= \int_0^{\infty} t^n f(t) dt$$

** clean up at home*

$$\text{Let, } \mathcal{L}\{t^2 \sin t\} = \frac{a}{s^2 + a^2} = F(s)$$

$$\mathcal{L}\{t^2 \sin t\} = (-1)^2 \frac{d^2}{ds^2} \left(\frac{a}{s^2 + a^2} \right)$$

$$= 1 \cdot \frac{d}{ds} \left\{ -a(s^2 + a^2)^{-2} \cdot 2s \right\}$$

$$= -2a \left(\frac{2s}{(s^2 + a^2)^2} + \frac{2(s^2 + a^2)^{-2} \cdot 2s^2}{(s^2 + a^2)^2} \right)$$

Inverse Transformation

* IP $F(s)$

Laplace:



Inverse:



$$f(t) = 1$$

$$L\{f(t)\} = L\{1\} = \frac{1}{s} = F(s)$$

$$L^{-1}\{F(s)\} = L^{-1}\left(\frac{1}{s}\right) = 1 = f(t)$$

~~$\frac{1}{s}$~~ $Y(s) =$

Linearity of Inverse

everything that works for L.T, works for Inverse.

Example 1: $Y(s) = \frac{2}{s}$

Ans. $L^{-1}\{Y(s)\} = L^{-1}\left\{\frac{2}{s}\right\} = 2L^{-1}\left\{\frac{1}{s}\right\}$

$$= 2 \cdot 1 = 2 \text{ (Ans.)}$$

$$\therefore y(t) = 2$$

Example 2: $Y(s) = \frac{3}{s-5}$

Ans. $L^{-1}\{Y(s)\} = L^{-1}\left\{\frac{3}{s-5}\right\} = 3L^{-1}\left\{\frac{1}{s-5}\right\}$

$$= 3e^{5t}$$

$$\therefore y(t) = 3e^{5t} \text{ (Ans.)}$$

Example 3: $Y(s) = \frac{6}{s^4} = \frac{3!}{s^{3+1}}$

Ans. $L^{-1}\{Y(s)\} = L^{-1}\left\{\frac{6}{s^4}\right\} = L^{-1}\left\{\frac{3!}{s^{3+1}}\right\}$

$$= t^3$$

$$\therefore y(t) = t^3 \text{ (Ans.)}$$

Example 4: $\gamma(s) = \frac{8}{s^3}$

$$L^{-1}\{\gamma(s)\} = L^{-1}\left\{\frac{4 \times 2!}{s^{2+1}}\right\} = 4 L^{-1}\left(\frac{2!}{s^{2+1}}\right)$$

$$= 4t^2$$

$$\therefore y(t) = 4t^2$$

Example 5: $\gamma(s) = \frac{4s+1}{s^2+9}$

$$L^{-1}\{\gamma(s)\} = L^{-1}$$

$$\gamma(s) = \frac{4s+1}{s^2+9} = \frac{4s}{s^2+9} + \frac{1}{s^2+9}$$

$$= 4 \left(\frac{s}{s^2+3^2} \right) + \left(\frac{3}{s^2+3^2} \right) \cdot \frac{1}{3}$$

$$\therefore L^{-1}\{\gamma(s)\} =$$

$$\therefore y(t) = 4 \cos 3t + \frac{1}{3} \sin 3t$$

Example 6: $Y(s) = \frac{4s+1}{s^2-9}$

Example 7: $Y(s) = -\frac{10}{(s+1)^3}$

$$\therefore Y(s) = -5 \left(\frac{2!}{(s+1)^{2+1}} \right)$$

$$\therefore L^{-1} \{ Y(s) \} = -5t^2 e^{-t}$$

Transforms of Derivative

$$L\{f'(t)\} = \int_0^{\infty} e^{-st} f'(t) dt \quad ; \quad \left[\int u v dt = u \int v dt - \int \left\{ \frac{d}{dt}(u) \int v dt \right\} dt \right]$$

$$= \left[e^{-st} \int f'(t) dt \right]_0^{\infty} - \int_0^{\infty} \left\{ \frac{d}{dt} (e^{-st}) \int f'(t) dt \right\} dt$$

$$= e^{-st} f(t) \Big|_0^{\infty} - \int_0^{\infty} -s e^{-st} f(t) dt$$

$$= 0 - f(0) + s \int_0^{\infty} e^{-st} f(t) dt$$

$$\therefore L\{f'(t)\} = sF(s) - f(0)$$

$$L\{f''(t)\} = \int_0^{\infty} e^{-st} f''(t) dt$$

$$= \left[e^{-st} \int f''(t) dt \right]_0^{\infty} - \int_0^{\infty} \left\{ \frac{d}{dt} (e^{-st}) \int f''(t) dt \right\} dt$$

$$= 0 - f'(0) - \int_0^{\infty} -s e^{-st} f'(t) dt$$

$$= s(sF(s) - f(0)) - f'(0)$$

$$= s^2 F(s) - s f(0) - f'(0)$$

$$\therefore L\{f'''(t)\} = s^3 F(s) - s^2 f(0) - s f'(0) - f''(0)$$

$$\therefore L\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$$

Consider:

$$y'' + 5y' + 6y = 0, \quad y(0) = 2, \quad y'(0) = 3$$

Taking Laplace Transformation on both the sides,

$$L\{y''\} + 5L\{y'\} + 6L\{y\} = L\{0\}, \quad [\text{Linearity}]$$

$$\text{on, } \{s^2 Y(s) - s y(0) - y'(0)\}$$

$$+ 5 \{s Y(s) - y(0)\} + 6 Y(s) = 0$$

$$\text{on, } s^2 Y(s) \{s^2 + 5s + 6\} - y(0) \{s - 5\} - y'(0) = 0$$

$$\text{on, } Y(s) \{s^2 + (s+3)(s+2)\} - y(0)(s-5) - y'(0) = 0$$

$$= 2(s-5) + 3 = 2s - 10 + 3 = 2s - 7$$

$$\therefore Y(s) = \frac{2s - 7}{s^2 + 5s + 6} = \frac{2s - 7}{(s+3)(s+2)}$$

Now, taking inverse Laplace,

$$L^{-1}\{Y(s)\} = L^{-1}\left\{\frac{2s - 7}{(s+3)(s+2)}\right\}$$

(Thumb rule is allowed)

$$\frac{2s + 13}{(s+3)(s+2)} = \frac{A}{(s+3)} + \frac{B}{(s+2)}$$

$$s = -2; B = 9$$

$$s = -3; A = 7, \quad B = 9$$

$$L^{-1}\{Y(s)\} = L^{-1}\left\{-\frac{7}{s+3}\right\} + L^{-1}\left\{\frac{9}{s+2}\right\}$$

$$\therefore y(t) = -7e^{-3t} + 9e^{-2t}$$

C-5(w-c)

12/05/24

Convolution Theorem

Example:

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{1\} = \frac{1}{s}$$

$$\mathcal{L}\{g(t)\} = \mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}$$

$$\therefore \mathcal{L}\{f(t)g(t)\} \neq \mathcal{L}\{f\} \mathcal{L}\{g\}$$

$$\mathcal{L}^{-1}\{H(s)\} = h(t)$$

$$= \int_0^t f(t-\tau)g(\tau) d\tau \quad \rightarrow \text{Convolution Integration}$$

$$= \int_0^t f(\tau)g(t-\tau) d\tau$$

[Proof in slide]

Example 2:

$$\text{Let, } f(t) = t$$

$$g(t) = \sin 2t$$

$$H(s) = F(s)G(s)$$

$$\therefore L^{-1}\{H(s)\} = h(t)$$

$$= \int_0^t (t - \varphi) \sin 2\varphi d\varphi$$

$$= t \int_0^t \sin 2\varphi d\varphi - \int_0^t \varphi \sin 2\varphi d\varphi$$

$$= -t \cdot \frac{\cos 2\varphi}{2} \Big|_0^t + \frac{\varphi \cos 2\varphi}{2} \Big|_0^t + \int_0^t 1 \left(-\frac{\cos 2\varphi}{2} \right) d\varphi$$

$$= -t \left(\frac{\cos 2t}{2} - \frac{1}{2} \right) + \frac{t \cos 2t}{2} - \frac{\sin 2\varphi}{4} \Big|_0^t$$

$$= \frac{t}{2} - \frac{\sin 2t}{4} \text{ (Ans.)}$$

Example 3 :

$$H(s) = \frac{2}{s^2(s-2)}$$

Let,

$$F(s) = \frac{2}{s^2} \quad f(t) = 2t$$

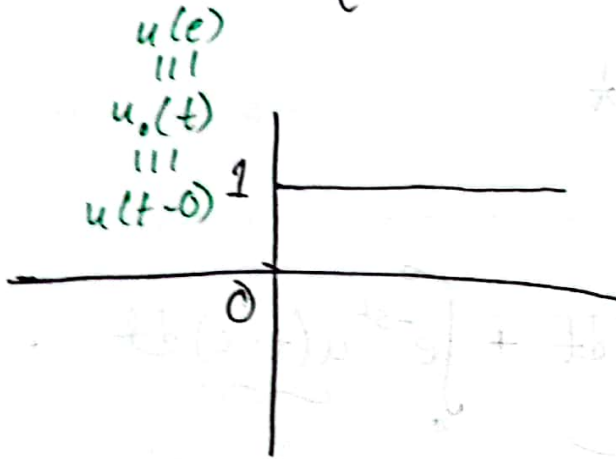
$$G(s) = \frac{1}{s-2} \quad g(t) = e^{2t}$$

$$\begin{aligned} \therefore L^{-1}\{H(s)\} &= \int_0^t 2(t-\tau) 2e^{2\tau} d\tau \\ &= 2 \int_0^t \tau e^{2\tau} d\tau - 2 \int_0^t 1 e^{2\tau} d\tau \\ &= 2t \int_0^t e^{2\tau} d\tau \\ &= 2t \cdot \frac{1}{2} e^{2\tau} \Big|_0^t - 2 \int_0^t \frac{1}{2} e^{2\tau} d\tau \quad 2 \left[\frac{\tau e^{2\tau}}{2} \right]_0^t \\ &\quad - 2 \int_0^t \frac{e^{2\tau}}{2} d\tau \\ &= te^{2t} - t - te^{2t} + \frac{e^{2\tau}}{4} \Big|_0^t \\ &= t + \frac{e^{2t}}{2} - \frac{1}{2} \end{aligned}$$

Heaviside's Unit Step

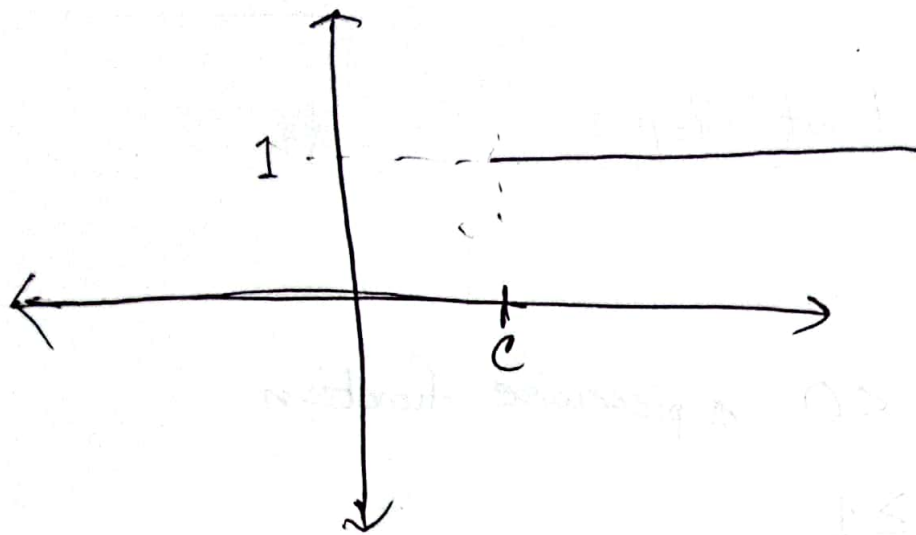
* Heaviside Function:

$$u(t) = \begin{cases} 0; & t < 0 \\ 1; & t \geq 0 \end{cases} \quad \text{a piecewise function}$$



$$u(t) = \begin{cases} 0; & t < 0 \\ 1; & t \geq 0 \end{cases} = 0 \cdot \{u(t-0)\} + 1 \cdot \{u(t-0)\} \\ = u(t)$$

$$\begin{aligned} \mathcal{L}\{u(t)\} &= \int_0^{\infty} e^{-st} u(t) dt \\ &= \int_0^{\infty} e^{-st} \cdot 1 dt; \quad [u(t) = 1 \text{ when } t \geq 0] \\ &= \frac{1}{s} \end{aligned}$$



hence,

$$u(t) = \begin{cases} 0; & t < c \\ 1; & t \geq c \end{cases}$$

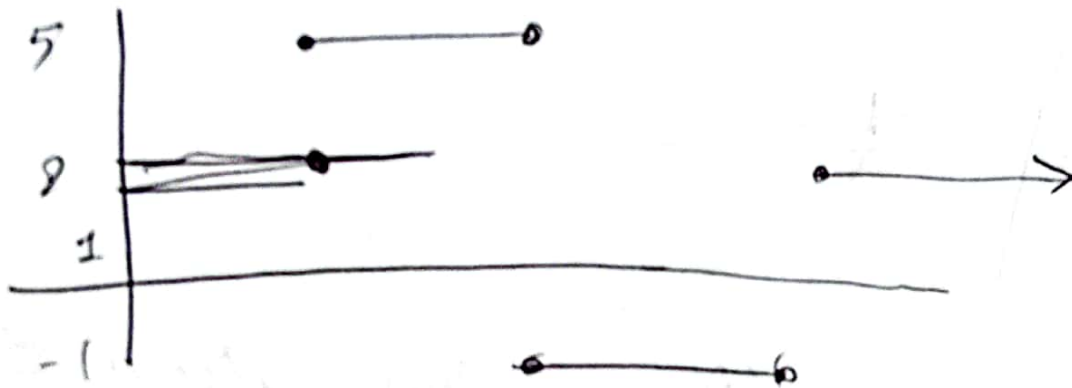
$$\mathcal{L}\{u(t-c)\} = \int_0^{\infty} e^{-st} u(t-c) dt$$

$$= \underbrace{\int_0^c e^{-st} \underbrace{u(t-c)}_0 dt}_0 + \int_c^{\infty} e^{-st} \underbrace{u(t-c)}_1 dt$$

$$= \int_c^{\infty} e^{-st} dt = \left[\frac{1}{s} e^{-st} \right]_c^{\infty} = \frac{0 - \left(-\frac{1}{s} e^{-ct}\right)}{s} = \frac{e^{-ct}}{s}$$

Example 4:

$$f(t) = \begin{cases} 2; & 0 \leq t < 4 = 2 \{u(t-0) - u(t-4)\} \\ 5; & 4 \leq t < 7 + 5 \{u(t-4) - u(t-7)\} \\ -1; & 7 \leq t < 9 + (-1) \{u(t-7) - u(t-9)\} \\ 1; & t \geq 9 + 1 \{u(t-9)\} \end{cases}$$



$$\therefore f(t) = 2u(t) + 3u(t-4) - 6u(t-7) + 2u(t-9)$$

$$\therefore \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$= 2\mathcal{L}\{u(t)\} + 3\mathcal{L}\{u_4(t)\} - 6\mathcal{L}\{u_7(t)\} + 2\mathcal{L}\{u_9(t)\}$$

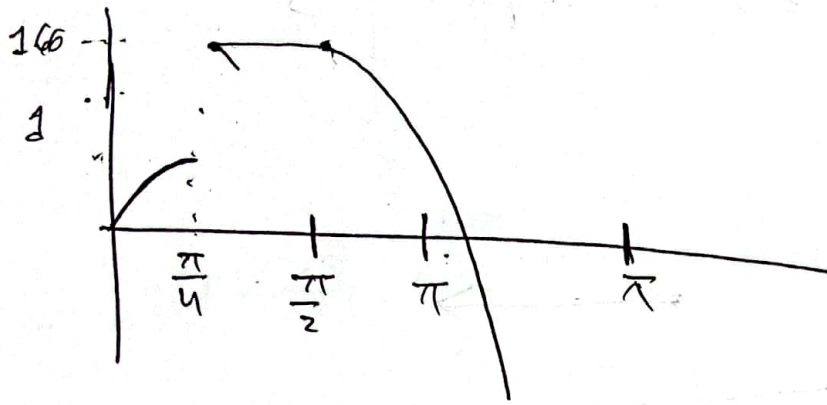
$$= 2 \cdot \frac{1}{s} + 3 \cdot \frac{e^{-4t}}{s} - 6 \cdot \frac{e^{-7t}}{s} + 2 \cdot \frac{e^{-9t}}{s}$$

$$= \frac{1}{s} (2 + 3e^{-4t} - 6e^{-7t} + 2e^{-9t})$$

Example 2:

1111

$$f(t) = \begin{cases} \sin t & ; 0 \leq t < \frac{\pi}{4} \\ \sin t + \cos(t - \frac{\pi}{4}) & ; t \geq \frac{\pi}{4} \end{cases}$$



$$\therefore f(t) = u_0(t) \sin t - u_{\frac{\pi}{4}}(t) \sin t + u_{\frac{\pi}{4}}(t) \sin t + u_{\frac{\pi}{4}}(t) \cos(t - \frac{\pi}{4})$$

$$= u_0(t) \sin t + u_{\frac{\pi}{4}}(t) \cos(t - \frac{\pi}{4})$$

$$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{u_0(t) \sin t\} + \mathcal{L}\{u_{\frac{\pi}{4}}(t) \cos(t - \frac{\pi}{4})\}$$

$$= \mathcal{L}\{\sin t\} + \mathcal{L}\{\cos(t - \frac{\pi}{4}) u(t - \frac{\pi}{4})\}$$

$$= \frac{1}{s^2 + 1} + e^{-\frac{s\pi}{4}} \cdot \frac{s}{s^2 + 1}$$

time shift

Complex VariableSolution of DE Using LT

$$\# \quad 2y'' + y' + 2y = g(t) ; y(0) = 0, y'(0) = 0$$

where,

$$g(t) = u_5(t) - u_{20}(t) = \begin{cases} 1, & 5 \leq t \leq 20 \\ 0, & 0 \leq t \leq 5 \text{ and } t \geq 20 \end{cases}$$

Here,

$$g(t) = 1 \{ u(t-5) - u(t-20) \}$$

$$= u_5(t) - u_{20}(t)$$

$$\therefore 2y'' + y' + 2y = u_5(t) - u_{20}(t)$$

$$\text{or, } 2\{y''\} + L\{y'\} + 2L\{y\} = L\{u_5(t)\} - L\{u_{20}(t)\}$$

$$\text{or, } 2\{s^2 Y(s) - sy(0) - y'(0)\} + \{sY(s) - y(0)\} = \frac{e^{-5s}}{s} - \frac{e^{-20s}}{s}$$

$$- 2y'(0) \quad 2Y(s) = \frac{e^{-5s}}{s} - \frac{e^{-20s}}{s}$$

$$\text{or, } Y(s) \{2s^2 + s + 2\} - y(0)(2s + 1) - y'(0) = \frac{1}{s}(e^{-5s} - e^{-20s})$$

$$\text{or, } (2s^2 + s + 2)Y(s) = \frac{e^{-5s} - e^{-20s}}{s}$$

$$\therefore Y(s) = \frac{e^{-5s} - e^{-20s}}{s(2s^2 + s + 2)}$$

Let, $\frac{1}{s(2s^2 + s + 2)} = \frac{A}{s} + \frac{Bs + C}{2s^2 + s + 2}$

$$\text{or, } \frac{1}{s(2s^2 + s + 2)} = A(2s^2 + s + 2) + (Bs + C)(s)$$

$$= 2As + A \cdot 2s^2 +$$

$$= s^2(2A + B) + s(A + C) + 2A$$

$$\therefore \begin{cases} 2A + B = 0 \\ A + C = 0 \end{cases} \quad \begin{cases} 2A = 1 \\ B = -\frac{1}{2} \\ C = -\frac{1}{2} \end{cases}$$

$$\therefore A = \frac{1}{2}$$

$\therefore$ we will get,

$$\begin{aligned} \therefore Y(s) &= e^{-5s} - e^{-20s} \left(\frac{1}{2s} + \frac{-s - \frac{1}{2}}{2s^2 + s + 2} \right) \\ &= (e^{-5s} - e^{-20s}) \left(\frac{1}{2s} - \frac{2s + 1}{4s^2 + 2s + 4} \right) \\ &= \frac{e^{-5s}}{2s} - \frac{e^{-20s}}{2s} - e^{-5s} \frac{2s + 1}{4s^2 + 2s + 4} + \frac{e^{-20s}(2s + 1)}{4s^2 + 2s + 4} \\ &= \frac{1}{2s}(e^{-5s} - e^{-20s}) - \frac{e^{-5s}(2s + 1)}{\{(2s)^2 + 2 \cdot 2s \cdot \frac{1}{2} + \frac{1^2}{4}\} - \frac{1}{4}} + \frac{e^{-20s}(2s + 1)}{\{(2s)^2 + 2s + \frac{1}{4}\} - \frac{1}{4}} \end{aligned}$$

$$= \frac{1}{2s} (e^{-5s} - e^{-20s}) - \frac{e^{-5s}(2s+1)}{2s}$$

* Multiple Equations = System of Equation

* Circuit related question will be in exam

$$0.01 \frac{di_2}{dt} + 5i_2 + 5i_3 = 100$$

$$0.0125 \frac{di_3}{dt} + 5i_2 + 5i_3 = 100$$

d

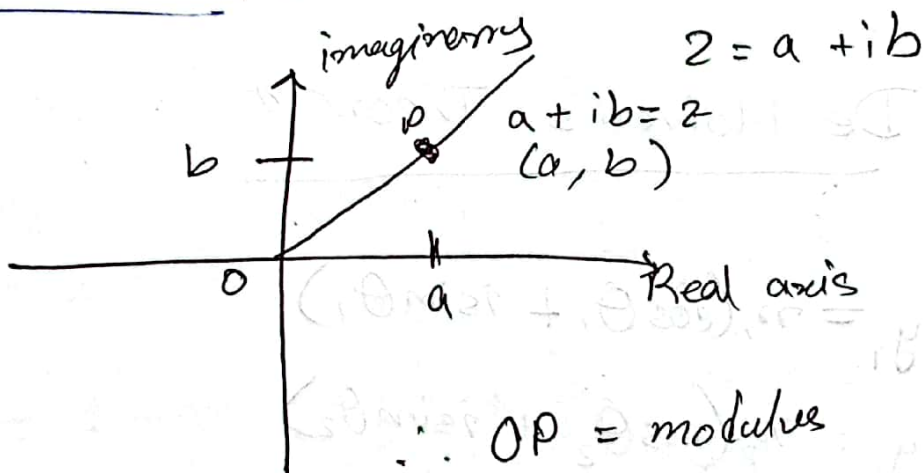
Complex Numbers System

$\sqrt{-1} = i$, $z = a + ib$

$\rightarrow$ Real part $\text{Re}(z)$
 $\rightarrow$ Imaginary part $\text{Im}(z)$

* If 2 complex numbers are equal, their real and imaginary numbers are also equal.

* Absolute value (distance from the origin)



$$\therefore OP = \sqrt{a^2 + b^2}$$

$$\therefore |a + ib| = \sqrt{a^2 + b^2}$$

* De Moivre (?) Theorem:

$$(\cos \theta + i \sin \theta)^n = \cos \theta + \cos \frac{\theta}{n} + i \sin \frac{\theta}{n}$$

$\omega^n = 1$

$$\text{or, } \omega = 1^{\frac{1}{n}} = n^{\frac{1}{n}} (\cos \theta + i \sin \theta)^{\frac{1}{n}}$$

$$= n^{\frac{1}{n}} \left(\cos \frac{\theta}{n} + i \sin \frac{\theta}{n} \right)$$

$$= n^{\frac{1}{n}} \left(\cos \frac{2k\pi\theta}{n} + i \sin \frac{2k\pi\theta}{n} \right); \quad \begin{matrix} \cos \theta = \cos(2n\pi + \theta) \\ \uparrow \\ k = 0, 1, \dots, n \end{matrix}$$

De Moivre's Theorem

$$\text{Let, } z_1 = x_1 + iy_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$$

$$\text{and, } z_2 = x_2 + iy_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$$

$$\therefore z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

$$\therefore \frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

$$\therefore z^n = r^n (\cos n\theta + i \sin n\theta)$$

Euler's Theorem

Using Maclaurin's Theorem,

$$\cos \theta x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\sin \theta x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$z = x + iy$$

$$= r \cos \theta + i r \sin \theta$$

$$= r (\cos \theta + i \sin \theta)$$

$$\therefore e^{ix} = 1 + ix + \frac{i^2 x^2}{2!} + \frac{i^3 x^3}{3!} + \dots$$

$$= 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \dots$$

$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + i \left(x - \frac{x^3}{3!} + \dots \right)$$

$$= \cos \theta - i \sin \theta$$

* Function: A relation between two sets of numbers.
under certain condition

$$f: A \rightarrow B; f(x) = x + 1$$

* Model is function

* Complex Numbers:

A complex function is a function f whose domain and range are subsets of $\mathbb{C}$

$$f: A \rightarrow B; A, B \subseteq \mathbb{C}$$

*Elementary Functions:

a. Polynomial: Exponent is positive integers

$$w = a_0 z^n + a_1 z^{n-1} + \dots + a_n = P(z)$$

b. Rational: Numerator & Denominator must be complex
(Den $\neq 0$)

$$w = \frac{P(z)}{Q(z)}$$

c. Exponential: Exponential is complex

$$a^z = e^{z \ln a}$$

d. Trigonometry: Trig function having complex

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\tan z = \frac{e^{iz} - e^{-iz}}{i(e^{iz} + e^{-iz})}$$

and all its friends.

e. Hyperbolic Function:

$$\sinh z = \frac{e^z - e^{-z}}{2}$$

$$\cosh z = \frac{e^z + e^{-z}}{2}$$

$$\tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

and all its friends.

f. Logarithmic Function:

$$w = \ln z = \ln r + i(\theta + 2k\pi); k = 0, \pm 1, \pm 2$$

g. Inverse Trigonometric Function:

$$\sin^{-1} z = \frac{1}{i} \ln (iz + \sqrt{1 - z^2})$$

$$\cos^{-1} z = \frac{1}{i} \ln \left(\frac{z}{i} + \sqrt{z^2 - 1} \right)$$

$$\tan^{-1} z = \frac{1}{2i} \ln \left(\frac{1 + iz}{1 - iz} \right)$$

n. Inverse Hyperbolic:

$$\sinh^{-1} z = \ln \left(z + \sqrt{z^2 + 1} \right)$$

$$\cosh^{-1} z = \ln \left(z + \sqrt{z^2 - 1} \right)$$

$$\tanh^{-1} z = \frac{1}{2} \ln \left(\frac{1+z}{1-z} \right)$$

Terminology:

Distance : $|z - z_0|$

Circle : $|z - z_0| = r$

Intension of Circle : $|z - z_0| < r$

Extension " " : $|z - z_0| > r$

Annulus : $r_1 < |z - z_0| < r_2$

Neighbourhood : All points in $|z - z_0| < r$

Boundary Point : Point not in intension and extension point

Open Set : Does not contain boundary points.

Closed set : Contains everything inside boundary points

Connected Set : If two sets ~~have~~ can be connected by a polygonal line they are connected set.

Domain : Open and connected set

Bounded region : Region enclosed by a circle.

Compact region : Closed and bounded region.

Limit Function of Complex Variable

$$f(z) \longrightarrow L$$

$$\text{when } z \longrightarrow L$$

$$\lim_{z \rightarrow a} f(z) = L$$

$$|f(z) - L| < \varepsilon ; [\varepsilon \text{ very small, close to } 0 \text{ } \overset{\text{positive numbers}}{\text{}}]$$

$$\text{if, } |z - a| < \delta ; [\delta \text{ " " " " " "}]$$

$$\text{and, } \delta = \delta(\varepsilon)$$

*Slide Math: DIY

+(03/07/24)

#Prove that

$$\lim_{\substack{z \rightarrow 1+i \\ z \rightarrow a}} (2+i)z = 1+3i ; \text{ Goal: } |z - a| < \delta \quad L = 1+3i$$

Let,

$$|f(z) - L| < \varepsilon$$

$$\text{or, } |(2+i)z - (1+3i)| < \varepsilon$$

$$\text{or, } |2+i| \times \left| z - \frac{1+3i}{2+i} \right| < \varepsilon ; [\text{Force common } |2+i|]$$

$$\text{or, } \sqrt{2^2 + 1^2} \times \left| z - \frac{(1+3i)(2-i)}{(2+i)(2-i)} \right| < \varepsilon$$

$$\text{or, } \sqrt{5} \times \left| z - \frac{2-i+6i-3i}{5} \right| < \varepsilon$$

Q. 12

$$\text{or, } \sqrt{5} \left| z - \frac{5+5i}{5} \right| < \cancel{\epsilon} \epsilon$$

$$\therefore |z - (1+i)| < \frac{\epsilon}{\sqrt{5}} = \delta; \left[\begin{array}{l} \text{Anything} \\ \epsilon \text{ is } \delta \end{array} \right] \text{ Any function of } \delta$$

(Proved)

* Continuity In Terms of Imaginary Numbers

* A real function is continuous when,

i) $f(x)$ at $x=a$ exists

ii) $\lim_{x \rightarrow a} f(x)$ exists

iii) $\lim_{x \rightarrow a} f(x) = f(a)$

* A complex number is continuous when,

$$|f(z) - f(z_0)| < \epsilon \text{ and } |z - z_0| < \delta$$

$\therefore f(z)$ is continuous at z_0 if,

$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$

Show that the function $f(z)$ defined by,

$$f(z) = \begin{cases} \frac{\operatorname{Re}(z)}{z} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$$

is not continuous at $z=0$

Here,

$z = 0$, $f(z)$ exists

For limit,

$$\lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \frac{\operatorname{Re}(z)}{z}$$

$$= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x}{x + iy}$$

Now/Along y -axis,

$$\begin{aligned} \lim_{y \rightarrow 0} \left| \lim_{x \rightarrow 0} \frac{x}{x + iy} \right| &= \lim_{y \rightarrow 0} \frac{0}{0 + iy} \\ &= \lim_{y \rightarrow 0} 0 \\ &= 0 \end{aligned}$$

Again/Along x -axis,

$$\begin{aligned} \lim_{x \rightarrow 0} \left| \lim_{y \rightarrow 0} \frac{x}{x + iy} \right| &= \lim_{x \rightarrow 0} \frac{x}{x + 0} \\ &= \lim_{x \rightarrow 0} \frac{x}{x} \\ &= 1 \end{aligned}$$

$\therefore$ The function has different value at $z=0$,
does not have limit at

$\therefore$ The function is not differentiable at $z=0$.

$\therefore$ " " " " continuous at $z=0$.
(Shown)

Discuss the continuity of $f(z)$ at origin,

$$f(z) = \begin{cases} \frac{\bar{z}}{z} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$$

Here,

$f(z)$ exists at $z=0$

~~$\therefore$ Along X-axis, Now,~~

$$\lim_{z \rightarrow 0} \frac{\bar{z}}{z} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x - iy}{x + iy}$$

$\therefore$ Along X-axis,

$$\lim_{x \rightarrow 0} \left| \lim_{y \rightarrow 0} \frac{x - iy}{x + iy} \right| = \lim_{x \rightarrow 0} \frac{x}{x} = 1$$

$\therefore$ Along Y-axis,

$$\lim_{y \rightarrow 0} \left| \lim_{x \rightarrow 0} \frac{x - iy}{x + iy} \right| = \lim_{y \rightarrow 0} \frac{-iy}{iy} = -1$$

- $\therefore$ The function has different values at the origin,
- $\therefore$ The function does not have limit at the origin
- $\therefore$ The function ~~does not~~ is not continuous at origin.

Example 4: (a) Prove that $f(z) = z^2$ is continuous at z_0

(b) Prove that, $f(z) = \begin{cases} z^2 & z \neq z_0 \\ z_0^2 & z = z_0 \end{cases}$, where $z_0 \neq 0$, is discontinuous at $z = z_0$

(a) (ϵ, δ)

$$|z^2 - z_0^2| < \epsilon$$

$$\text{or, } |z + z_0||z - z_0| < \epsilon$$

$$\text{or, } |z - z_0| < \frac{\epsilon}{|z + z_0|}$$

$$\therefore |z - z_0| < \frac{\epsilon}{\sqrt{(x+x_0)^2 + (y+y_0)^2}} = \delta$$

$\therefore f(z)$ is continuous at $z = z_0$.

#Ex-5.

$$f(z) = \frac{3z^4 - 2z^3 + 8z^2 - 2z + 5}{z - i}$$

$$= \frac{(3z^4 - 2z^3 + 8z^2 - 2z + 5)(z + i)}{(z - i)(z + i)}$$

$$= \frac{(3z^4 - 2z^3 + 8z^2 - 2z + 5)(z + i)}{z^2 + 1}$$

$$\therefore f(i) = \frac{(3 + 2i - 8 - 2i + 5)(2i)}{z^2 + 1}$$

$$= \frac{3z^5 - 2z^4 + 8z^3 - 2z^2 + 5z + 3z^4i - 2z^3i + 8z^2i - 2zi + 5i}{z^2 + 1}$$

Do at home

$$= \frac{(z^2 + 1)(3z^3 - 2z^2 + 5z + 2i)}{z^2 + 1}$$

$$= \frac{3z^5 - 2z^4 + 8z^3 - 2z^2 + 5z + 3z^4i - 2z^3i + 8z^2i - 2zi + 5i}{z^2 + 1}$$

=

$$= \frac{(z^2 + 1)(3z^3 - 2z^2 + 5z + 2i)}{(z^2 + 1)}$$

$$= \frac{3z^5 + 3z^3 + 5z^2 + 5z - 2z^4 - 2z^2 + 3z^4i + 3z^2i + 5z^2i + 5i}{z^2 + 1}$$

$$= \frac{(z^2 + 1)3z^3 + (z^2 + 1)5z + (z^2 + 1)(-2z) + (z^2 + 1)(3z^2i) + (z^2 + 1)5i}{z^2 + 1}$$

Complex Functions as Mapping

[Check Slide + Book]

$z = u + iv$

$$= \frac{(z^2 + 1)(3z^3 + 5z - 2z^2 + 3iz^2 - 2z + 5i)}{(z^2 + 1)}$$

$$= 3z^3 + 5z - 2z^2 + 3iz^2 - 2z + 5i$$

$$= (z + i)(3z^2 - 2z + 5)$$

$$\therefore \lim_{z \rightarrow i} f(z) = \lim_{z \rightarrow i} (z + i)(3z^2 - 2z + 5)$$

$$= 2i(-3 - 2i + 5)$$

$$= 4 - 4i$$

$\therefore$ The limit exists at $z = i$

$\therefore$ The $f(z)$ is continuous at $z = i$

(Ans)

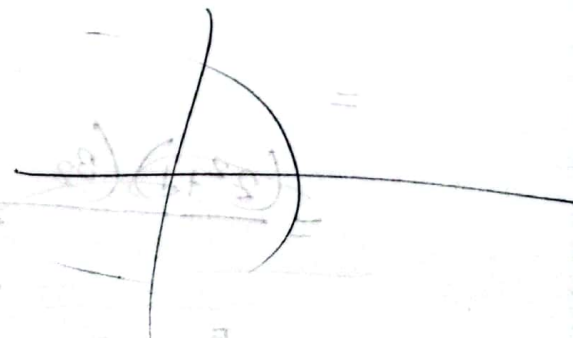
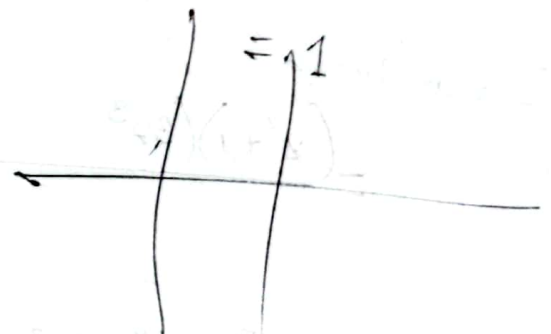
$$(i)^2 = 1 + 2iy + y^2$$

$$= (1 - y^2) + (2y) \cdot i$$

$$(z, f(z))$$

$$u = 2y \quad \therefore y = \frac{u}{2}$$

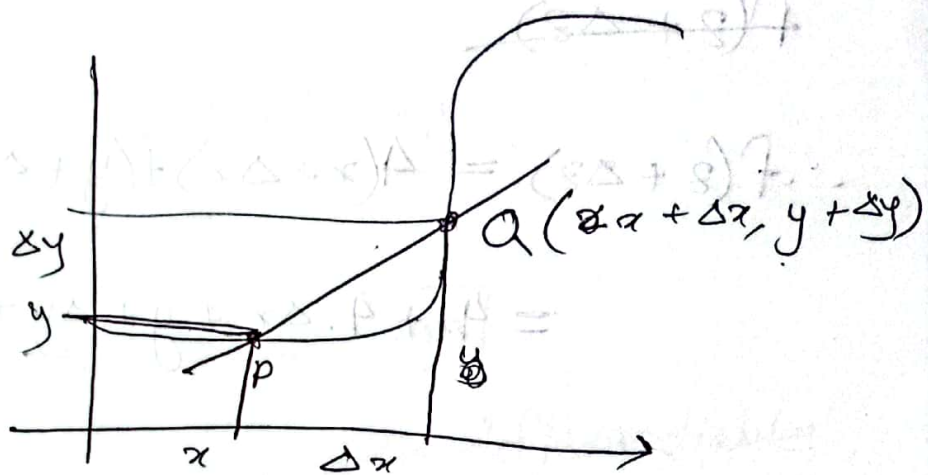
$$\therefore u = 1 - \frac{u^2}{4}$$



Analytic Function

* If a function is differentiable at any point, then the function will be analytic at that point.

$$y = f(x)$$



$$\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$w + \Delta w = f(z + \Delta z) - f(z_0)$$

$$\Delta w = f(z + \Delta z) - f(z_0)$$

$$\frac{\Delta w}{\Delta z} = \frac{f(z + \Delta z) - f(z_0)}{\Delta z}$$

$$\lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z_0)}{\Delta z}$$

variation

$f(z) = 4x + y + i(4y - x)$; discuss $\frac{df}{dz}$

$$\frac{df}{dz} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

~~$f(z + \Delta z)$~~

$$\begin{aligned}\therefore f(z + \Delta z) &= 4(x + \Delta x) + (y + \Delta y) + i\{4(y + \Delta y) - (x + \Delta x)\} \\ &= 4x + 4\Delta x + y + \Delta y + i(4y + 4\Delta y - x - \Delta x)\end{aligned}$$

$$\therefore f(z + \Delta z) - f(z) = 4\Delta x + \Delta y + i(4\Delta y - \Delta x)$$

$$\therefore \frac{df}{dz} = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{4\Delta x + \Delta y + i(4\Delta y - \Delta x)}{\Delta x + i\Delta y}$$

Along X-axis, (put $\Delta y = 0$)

$$\lim_{\Delta x \rightarrow 0} \frac{df}{dz} = \lim_{\Delta x \rightarrow 0} \frac{4\Delta x - i\Delta x}{\Delta x} = 4 - i$$

Along Y-axis, (put $\Delta x = 0$)

$$\begin{aligned}\frac{df}{dz} &= \lim_{\Delta y \rightarrow 0} \frac{\Delta y + 4i\Delta y}{i\Delta y} = \frac{1 + 4i}{i} \\ &= \frac{i - 4}{-1} \\ &= 4 - i\end{aligned}$$

Take Along $y = mx$,

$$\lim_{\substack{\Delta z \rightarrow 0 \\ y \rightarrow mx}} \frac{df}{dz} = \lim_{\Delta x \rightarrow 0} \frac{4\Delta x + m\Delta x + i(4m\Delta x - \Delta x)}{\Delta x + i\Delta x m}; [\Delta y = \Delta x]$$

$$= \lim_{\Delta x \rightarrow 0} \frac{4 + m + i(4m - 1)}{1 + im}$$

$$= \frac{4(1 + im) + i(4m - 1)(1 + im)}{1 + im}$$

$$= 4 - i; [\text{free from } m]$$

~~$\therefore$ It is not dependent on m , it is differentiable~~

~~$\therefore$ It is~~ all different path $\frac{df}{dz} = (4 - i)$ which is unique.

$\therefore$ Along all different path $\frac{df}{dz} = (4 - i)$ which is unique.

$\therefore f(z)$ is differentiable at point z

$f(z) = \begin{cases} \frac{z^3 y (y - iz)}{x^6 + y^2} ; z \neq 0 \\ 0 ; z = 0 \end{cases}$, then discuss $\frac{df}{dz}$ at $z=0$

$$\therefore \frac{df}{dz} = \lim_{z \rightarrow 0} \frac{f(z+z_0) - f(z_0)}{z_0}$$

$$= \lim_{\substack{z \rightarrow 0 \\ y \rightarrow 0}} \frac{\frac{z^3 y (y - iz)}{x^6 + y^2} - 0}{z + iy}$$

$$= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{z^3 y (y - iz)}{(x^6 + y^2)(z + iy)}$$

Along $y = mx$,

$$\lim_{x \rightarrow 0} \frac{x^3 (mx) (mx - iz)}{(x^6 + m^2 x^2) (x + imx)} = \lim_{x \rightarrow 0} \frac{mx^2 (m - i)}{(1 + im)(x^4 + m^2)}$$

$$= \lim_{x \rightarrow 0} 0$$

$$= \lim_{x \rightarrow 0} \frac{(mx^4)(mx - iz)}{(x^6 + imx^7 + m^2 x^3 + im^2 x^3)}$$

$$= \lim_{x \rightarrow 0} \frac{m^2 x^4 - imx^5}{x^6 + imx^7 + m^2 x^3 + im^2 x^3}$$

Along $y = x^3$,

$$\lim_{x \rightarrow 0} \frac{x^3 \cdot x^3 (x^3 - ix)}{(x^6 + x^6)(x + ix^3)} = \lim_{x \rightarrow 0} \frac{x^6 (x^3 - ix)}{(2x^6)(x + ix^3)}$$

$$= \lim_{x \rightarrow 0} \frac{-x^2 - i}{2(1 + ix^2)} = \frac{-i}{2}$$

$\therefore$ Along different axis, value is different

$\therefore$ It is not differentiable

Analytic at a Pointand neighbourhood $\rightarrow$ left limit, right limit existsDefⁿ: If differentiable at z_0 , it is analytic at that point.If every point ~~at~~ is differentiable, it is analytic at a domainHolomorphic: Within a domain if a function is analytic, it is holomorphic (synonym of analytic)Entire Function: Analytic at every point z in complex planeTheorem:

- (i) A polynomial function $p(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0$, where n is a non negative integer, is an entire function.
- (ii) A rational function $f(z) = \frac{p(z)}{q(z)}$ is analytic in any domain contain no point where $q(z_0) = 0$

Singular Point: If a function is not analytic at ^{any} point, that point is a singular point.

L'Hôpital :

For any function: $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$

where $f(z_0) = g(z_0) = 0$ but $g'(z_0) \neq 0$,

$$\text{then, } \lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$$

$$\# f(z) = z^2 - 4z + 5$$
$$g(z) = z^3 - z - 10i$$

$$\therefore f(2+i) = (2+i)^2 - 4(2+i) + 5$$
$$= 3 - 2i - 8 - 4i + 5$$

$$= 0$$

$$\therefore g(2+i) = (2+i)^3 - (2+i) - 10i$$
$$= 0$$

$$\therefore \lim_{z \rightarrow 2+i} \frac{f(2+i)}{g(2+i)} = \frac{0}{0}$$

$$\therefore \lim_{z \rightarrow 2+i} \frac{f(2+i)}{g(2+i)} = \frac{f'(2+i)}{g'(2+i)}$$
$$= \frac{2i}{9-6i}$$

$$\therefore f'(z) = 2z - 4$$
$$\therefore f'(2+i) = 4 + 2i - 4$$
$$= 2i$$
$$\therefore g'(z) = 3z^2 - 1$$
$$= 3(2+i)^2 - 1$$
$$= 9 - 6i - 1$$
$$= 8 - 6i \neq 0$$
$$= 8 -$$

#27. $f(z) = \frac{iz^2 - 2z}{3z + 1 - i}$

Given that,

$$f(z) = \frac{iz^2 - 2z}{3z + 1 - i}$$

If, $3z + 1 - i = 0$,

$$\therefore z = \frac{i-1}{3}$$

$\therefore \frac{i-1}{3}$ is a singular point of $f(z)$

$\therefore$ at $z = \frac{i-1}{3}$, $f(z)$ is not analytic.

Cauchy-Riemann Equation

If $f(z) = u(x, y) + iv(x, y)$ and f is differentiable at a point $z = x + iy$, then at z the first-order partial derivatives of u and v exist and satisfy the Cauchy-Riemann equations,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

+ Check proof & verification

If a function does not satisfy C-R equations, then the function is not analytic.

Necessary Condition for C-R:

$f(z)$ is analytic if

$$(i) \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad (ii) \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \text{ provided } \frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial y} \text{ exist}$$

Sufficient:

$$(i) \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$(ii) \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y} \text{ continuous}$$

C-R in Polar Form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} ; \quad \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$$

show that $w = e^z$ is analytic in complex plane. Hence

find $\frac{dw}{dz}$.

$$u = e^x \cos y, \quad v = e^x \sin y$$

$$\therefore \frac{\partial u}{\partial x} = \cancel{e^x \cos y} + \cancel{e^x \sin y} e^x \cos y$$

$$\therefore \frac{\partial u}{\partial y} = -e^x \sin y$$

$$\therefore \frac{\partial v}{\partial x} = e^x \sin y$$

$$\therefore \frac{\partial v}{\partial y} = e^x \cos y$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$\therefore$ CR equation is satisfied

$\therefore w$ is an analytic function.

$$\therefore \frac{dw}{dz} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$\swarrow$
 $w = e^z$

$\underbrace{\hspace{10em}}$
with respect to x and y

$$= e^x \cos y + i e^x \sin y$$

$$= e^x (\cos y + i \sin y)$$

$$= e^x \cdot e^{iy}$$

$$= e^{x+iy}$$

$$= e^z$$

Harmonic Function:

If $f(z) = u + iv$

where,

$f(z)$ is analytic

$$\text{and } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0,$$

then $f(z)$ is called harmonic.

∴ If $f(z)$ is analytic and harmonic, then u and v are ^{Conjugate} harmonic of each other.

$u = x^2 - y^2$ and $v = \frac{y}{x^2 + y^2}$ are harmonic but not conjugate harmonic.

$$\therefore \left. \begin{aligned} \frac{\partial u}{\partial x} &= 2x \\ \frac{\partial^2 u}{\partial x^2} &= 2 \end{aligned} \right| \left. \begin{aligned} \frac{\partial v}{\partial y} &= -2y \\ \frac{\partial^2 v}{\partial y^2} &= -2 \end{aligned} \right|$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\begin{aligned} \frac{\partial v}{\partial x} &= \frac{y}{x^2 + y^2} \\ &= \frac{0 \cdot (x^2 + y^2) - y \cdot 2x}{(x^2 + y^2)^2} \\ &= \frac{-(2xy)}{(x^2 + y^2)^2} \\ \therefore \frac{\partial^2 v}{\partial x^2} &= \frac{(x^2 + y^2)^2 (-2y) - 2(-2xy)(x^2 + y^2)}{(x^2 + y^2)^4} \end{aligned}$$

$$\frac{\partial v}{\partial y} = \frac{2xy}{(x^2+y^2)^2}$$

$$\frac{\partial^2 v}{\partial y^2} = \frac{2x^2 - 2y^2}{(x^2+y^2)^3}$$

$$\frac{\partial v}{\partial y} = \frac{x^2 + y^2 - 2y^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 v}{\partial y^2} = \frac{(x^2+y^2)^2 \cdot (-2) + (x^2+y^2)(2y)}{(x^2+y^2)^4}$$

$$\therefore \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{(x^2+y^2) \cdot (-2y) - 4(x^2+y^2)(x^2-y^2)y - (x^2+y^2)(-2x)}{(x^2+y^2)^4}$$

$$= 0$$

$\therefore$ They are harmonic

Here σ

$$\frac{\partial u}{\partial x} = 2x \neq \frac{\partial v}{\partial y}$$

$\therefore$ ~~$f(z)$ is~~ They are not conjugate harmonics.

Milne Thompson Method

$$z = x + iy, \quad \bar{z} = x - iy$$

$$\therefore x = \frac{z + \bar{z}}{2}, \quad y = \frac{z - \bar{z}}{2i}$$

$$\begin{aligned} \therefore f(z) &= u(x, y) + iv(x, y) \\ &= u\left(\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2i}\right) + iv\left(\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2i}\right) \end{aligned}$$

If $z = \bar{z}$,

then $f(z) = u(z, 0) + iv(z, 0)$; [this follows CR]

Case-1: If u is given,

step 1: Find $\frac{\partial u}{\partial x}$ and equate it to $\phi_1(x, y)$

" 2: Find $\frac{\partial u}{\partial y}$ " " " $\phi_2(x, y)$

" 3: Replace x with z and y with 0 to get $\phi_1(z, 0)$

" 4: " " z " y " 0 " " $\phi_2(z, 0)$

" 5: Find $f(z)$ by the formula $f(z) = \int \phi_1(z, 0) dz + i \int \phi_2(z, 0) dz$

Case 2: When v is given.

Step 1: Find $\frac{\partial v}{\partial x}$ and equate it to $\psi_2(x, y)$

" 2: " $\frac{\partial v}{\partial y}$ " " " $\psi_1(x, y)$

" 3: Replace x with z and y with 0 to get $\psi_1(z, 0)$
" $\psi_2(z, 0)$

" 4: " " " "

" 5: Find $f(z)$ using $f(z) = \int \{ \psi_1(z, 0) + i \psi_2(z, 0) \} dz + C$

Slide math

[Ex. 10.1] $\int_C \frac{1}{z} dz$ for a clockwise circle



Definite Integral

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$$

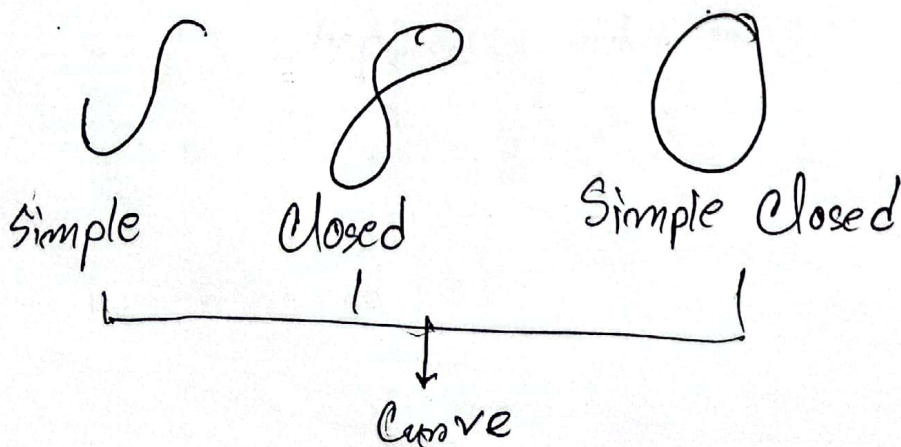
$$\int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(x_k^*) \Delta x_k \rightarrow R$$

$$y = f(x), \quad x = x(t)$$

Parameters

(independent variable of
a dependent variable)

t is a parameter of y [NOT x]



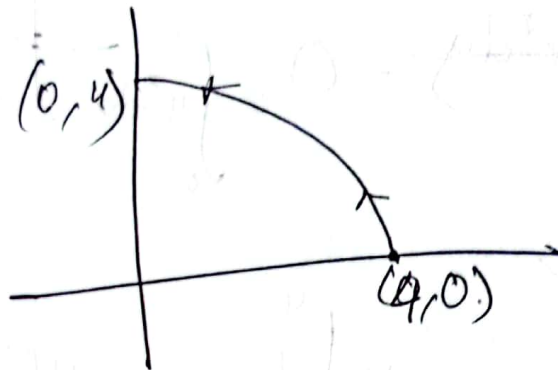
(a) $\int_C xy^2 dx$ (b) $\int_C xy^2 dy$ (c) $\int_C xy^2 ds$

Where C is a quarter circle defined by

$x = 4 \cos t$, $y = 4 \sin t$; $0 \leq t \leq \frac{\pi}{2}$

$x = 4 \cos t$
 $y = 4 \sin t$

$\therefore x^2 + y^2 = 16$



(a) $\int_C xy^2 dx$ $\left\{ \begin{array}{l} x = 4 \cos t \\ \therefore dx = -4 \sin t dt \end{array} \right.$

$= \int_0^{\frac{\pi}{2}} (4 \cos t) (4 \sin t)^2 (-4 \sin t dt)$

$= \int_0^{\frac{\pi}{2}} 4 \cos t (16 \sin^2 t) (-4 \sin t dt)$

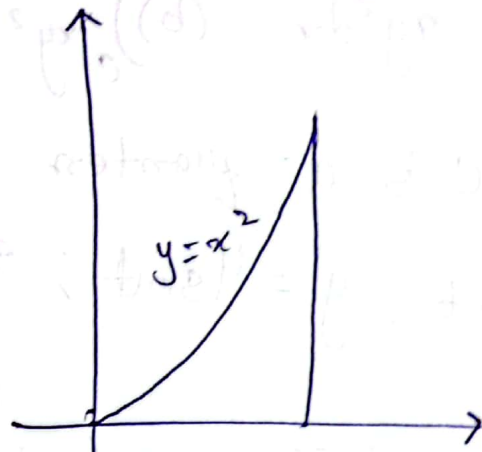
$= - \int_0^{\frac{\pi}{2}} 256 \sin^3 t \cos t dt = - \int_0^{\frac{\pi}{2}} 256 \sin^2 t \times \cos t \times \frac{d(\sin t)}{\cos t}$

$= -256 \left[\frac{1}{4} \sin^4 t \right]_0^{\frac{\pi}{2}}$

$$\# \oint_C y^2 dx - x^2 dy$$

Along x -axis,

$$I_1 = \int_0^2 y^2 dx - x^2 dy = 0 = \int_0^2 y^2 dx = 0$$



Along $x=2$,

$$I_2 = \int_0^4 -x^2 dy = -4 \int_0^4 dy = -4[y]_0^4 = -16$$

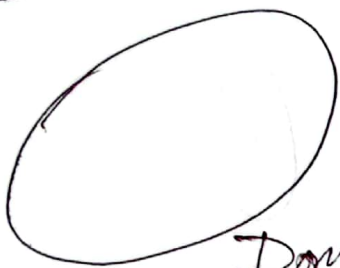
Along $y=x^2$,

$$\int_u^0 y^2 dx - x^2 dy \quad \left| \begin{array}{l} y^2 = x^2 \\ \therefore y^2 = x^4 \\ \therefore dy = 2x dx \end{array} \right| \begin{array}{l} y = x^2 \\ \therefore dy = 2x dx \end{array}$$

$$= \int_u^0 x^4 dx - x^2 (2x dx)$$

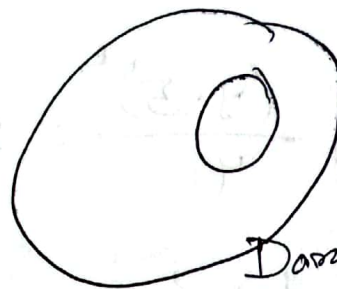
Cauchy Integral Theorem and Cauchy Integral Theorem Formula

Simply:



Domain has no holes
~~Q. Fully~~

Multiply:



Domain with holes
undefined

Cauchy Theorem:

$$\oint_C f(z) dz = 0$$

$C \rightarrow$ curve

* Contour:
closed curve

(i) Function f is analytic

(ii) " " " simply

(iii) f' is continuous in D then

$\rightarrow$ update: Cauchy Goursat said (iii) is not necessary

$\oint_C \frac{dz}{z^2}$ when ~~contour~~ contour C is ellipse
 $(x-2)^2 + \frac{(y-5)^2}{4} = 1$

Here, $f(z) = \frac{1}{z^2}$

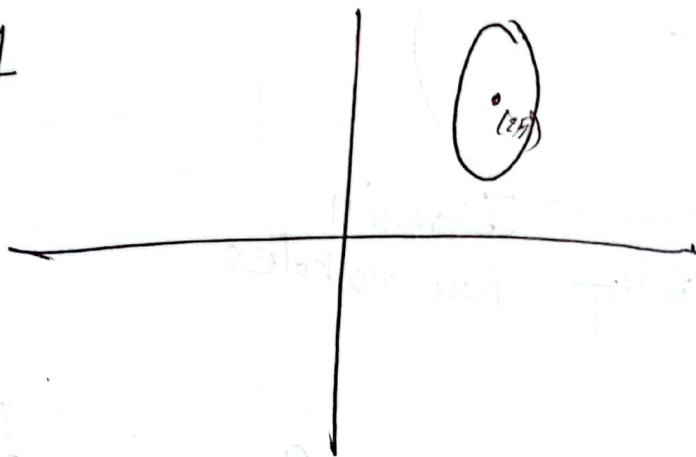
$\therefore$ At $z=0$, $f(z) = \frac{1}{z^2}$ is not analytic

Now,

$$\frac{(x-2)^2}{1} + \frac{(y-5)^2}{4} = 1$$

$$\therefore a = 1, b = 2$$

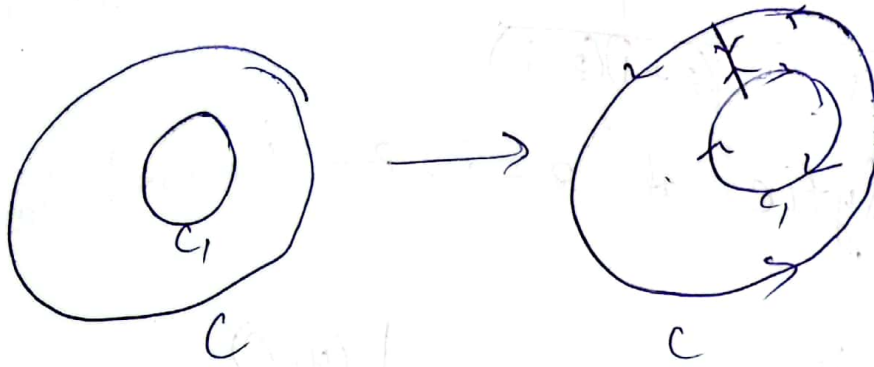
$$\therefore b > a$$



$\therefore z = f(z)$ is analytic in the curve

$$\therefore \oint_C \frac{dz}{z^2} = 0 \quad (\text{Ans})$$

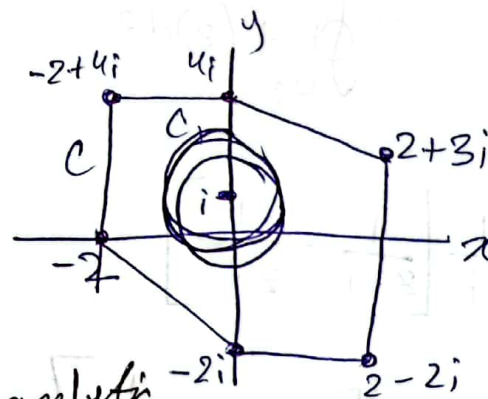
* Cauchy - Goursat Theorem for multiply connected



$$\therefore \oint_C f(z) dz = \oint_{C_1} f(z) dz + \left[\oint_{C_2} f(z) dz \dots \dots \dots \right]$$

if exists

* $\oint_C \frac{dz}{z-i}$ where C is



$$f(z) = \frac{1}{z-i}$$

at $z=i$, $f(z)$ is not analytic

$$= (0, 1)$$

$$\begin{aligned} \therefore \oint_C \frac{dz}{z-i} &= \oint_{C_1} f(z) dz \\ &= \oint_{C_1} \frac{ie^{i\theta} d\theta}{z-i} \\ &= \oint_0^{2\pi} \frac{ie^{i\theta}}{e^{i\theta}} d\theta = i \int_0^{2\pi} d\theta = 2\pi i \quad (\text{Ans}) \end{aligned}$$

$|z-i| = 1$
 or, $z = i + e^{i\theta}$
 or, $dz = ie^{i\theta} d\theta$

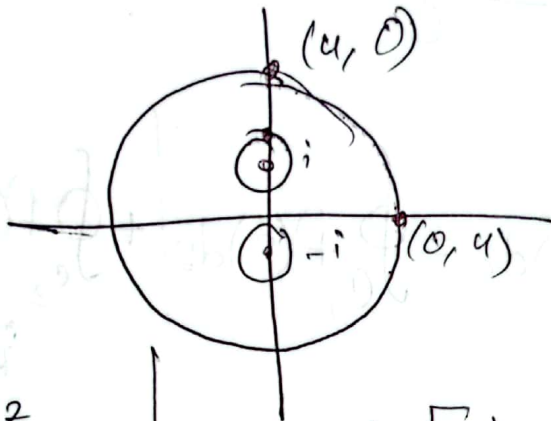
Example 3:

$$f(z) = \frac{1}{z^2 + 1} = \frac{1}{(z+i)(z-i)}$$

$\therefore$ Not analytic at $z = i, z = -i$

Here, $|z| = 4$ is

$$\therefore \oint_C f(z) dz = \oint_{C_1} f(z) dz + \oint_{C_2} f(z) dz$$



$$f(z) = \frac{1}{2i} \left[\frac{1}{z-i} - \frac{1}{z+i} \right]$$

$$= \oint_{C_1} \frac{1}{2i} \left[\frac{1}{z-i} - \frac{1}{z+i} \right]$$

$$+ \oint_{C_2} \frac{1}{2i} \left[\frac{1}{z-i} - \frac{1}{z+i} \right]$$

$$= \oint_{C_1} \frac{1}{2i} \times \frac{1}{z-i} - \oint_{C_1} \frac{1}{2i} \times \frac{1}{z+i} + \oint_{C_2} \frac{1}{2i} \times \frac{1}{z-i}$$

$$- \oint_{C_2} \frac{1}{2i} \times \frac{1}{z+i}$$

$$= 2\pi i - 0 + 0 - 2\pi i$$

$$= 0 \text{ (Ans)}$$

Cauchy's Integral Formula:

$$\text{1st theorem: } f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$$

point where

$$\text{2nd theorem: } f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad \text{not analytic}$$